

Covertypes Forest Cover Analysis

DSAA2011 machine learning project

581,012

rows

54

features

7

cover classes

0.845

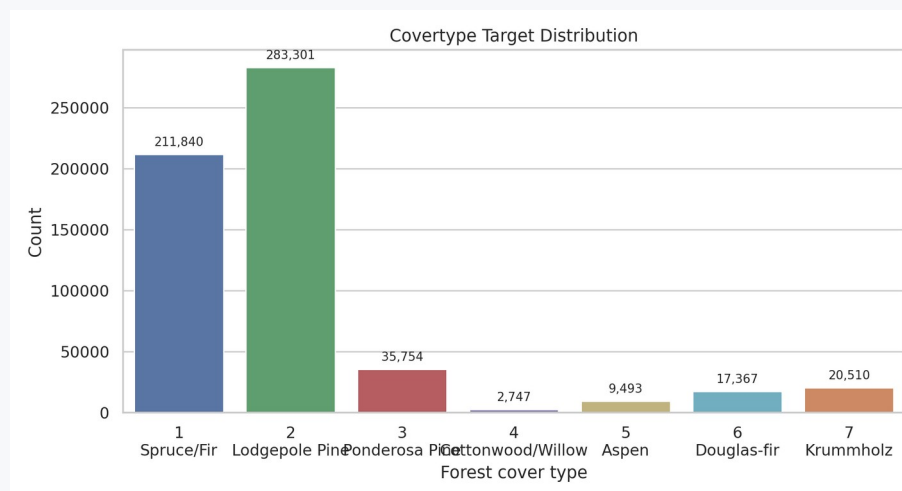
best macro-F1

0.902

best accuracy

0.991

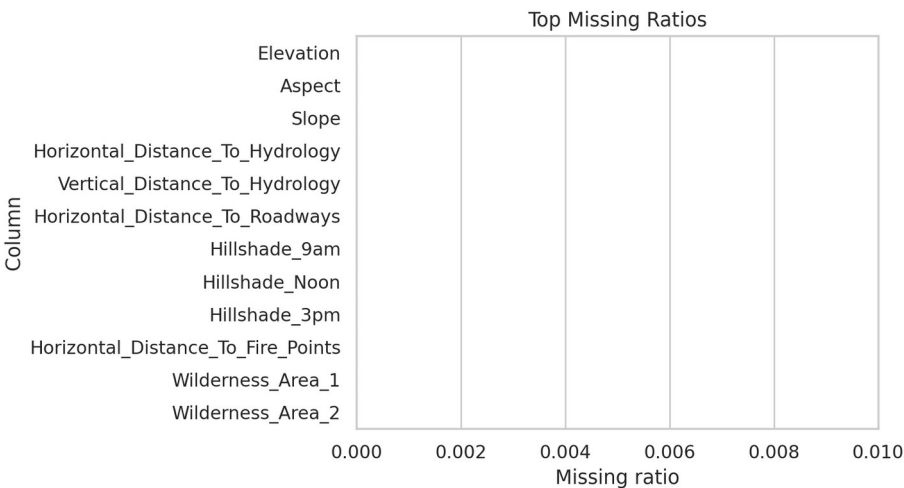
macro AUC



- Natural clusters reveal terrain structure but do not recover the seven labels.
- Wilderness Area analysis adds a regional robustness view.
- Random Forest is the best classifier on the stratified test set.
- Elevation, road/fire/hydrology distances, hillshade, soil context, and region explain errors.

Dataset and Preprocessing

UCI Coverture: cartographic measurements with a seven-class forest cover target

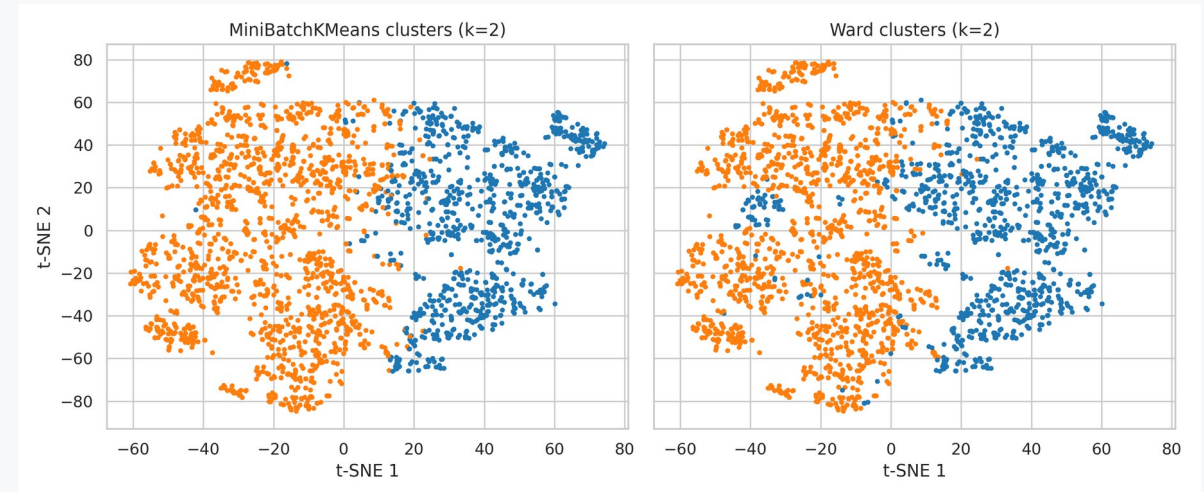
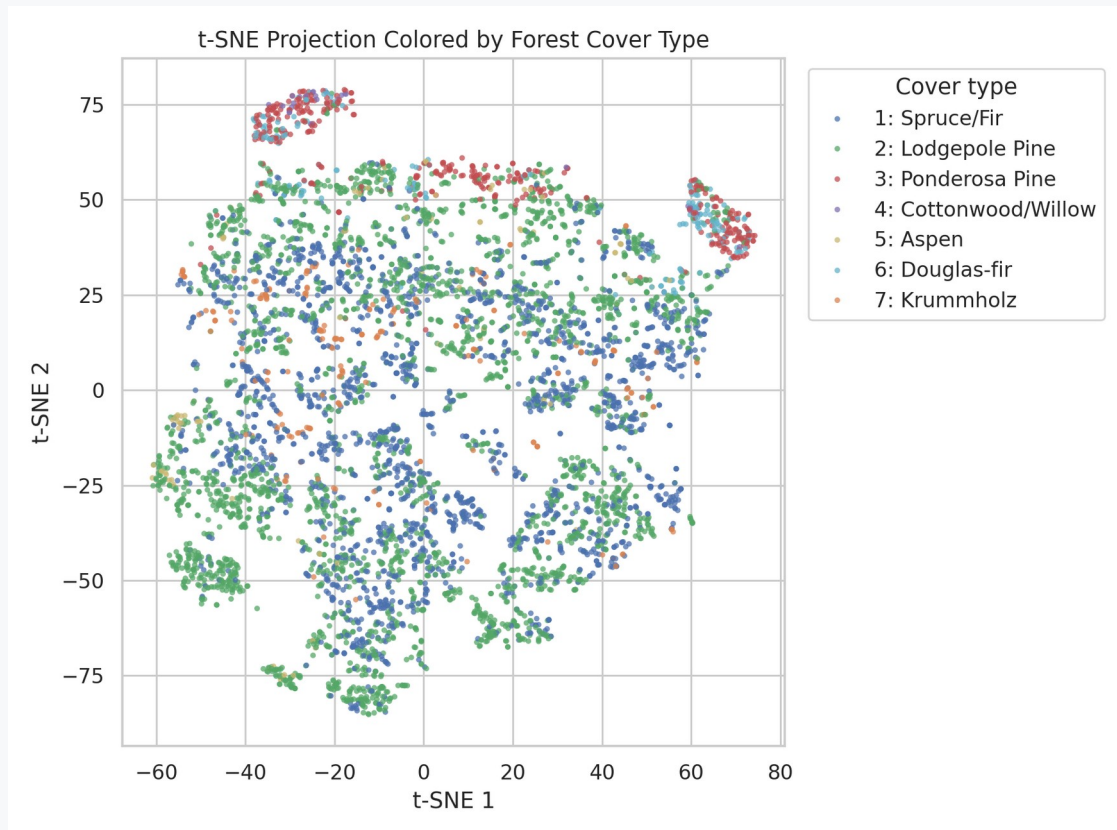


- 1 Spruce/Fir: 211,840 (36.5%)
- 2 Lodgepole Pine: 283,301 (48.8%)
- 3 Ponderosa Pine: 35,754 (6.2%)
- 4 Cottonwood/Willow: 2,747 (0.5%)
- 5 Aspen: 9,493 (1.6%)
- 6 Douglas-fir: 17,367 (3.0%)
- 7 Krummholz: 20,510 (3.5%)

- No missing values after parsing; maximum missing ratio is 0.000.
- 10 continuous terrain features are standardized for linear and distance-based methods.
- 4 wilderness and 40 soil indicators are already one-hot encoded.
- Large-scale steps use stratified samples to preserve class proportions while keeping runtime reproducible.

Visualization and Clustering

t-SNE shows partial local structure; clusters are not the same as cover labels



MiniBatchKMeans: k=2, silhouette=0.191, ARI target=0.003, ARI wild=0.039

Agglomerative Ward: k=2, silhouette=0.153, ARI target=0.004, ARI wild=0.013

- Near-zero ARI versus Cover_Type only means the natural partitions do not reproduce the supervised labels.
- ARI versus Wilderness is also modest, so clusters are better read as coarse terrain/region mixtures.

Simple Prediction Models

Logistic regression is a linear baseline; decision tree captures nonlinear splits

Logistic Regression: Confusion Matrices																							
Train								Test								Full							
Actual	1	2	3	4	5	6	7	Actual	1	2	3	4	5	6	7	Actual	1	2	3	4	5	6	7
	20291	5367	11	0	1100	150	3708		8594	2307	5	0	477	74	1669		13933	37468	88	0	7526	1050	26375
	9390	21599	702	100	7317	1433	417		3973	9312	320	37	3137	617	158		64229	149684	4836	643	51146	10039	2724
	0	39	2565	1320	283	962	0		0	13	1096	557	118	431	0		0	225	17719	9244	1913	6653	0
	0	0	13	381	0	3	0		0	0	4	162	0	4	0		0	0	113	2594	0	40	0
	9	210	30	0	1073	51	0		1	92	26	0	443	26	0		57	1466	257	0	7354	359	0
	0	6	498	505	116	1386	0		0	9	191	222	53	601	0		0	96	3314	3538	789	9630	0
Predicted								Predicted								Predicted							
1	20291	9390	0	0	9	333	2620	1	8594	3973	0	0	1	139	2105	1	13933	64229	0	0	62	0	18321
2	5367	21599	210	6	0	0	0	2	2307	9312	0	9	0	0	1	21	37468	149684	1	21	0	0	0
3	11	702	30	0	0	3	0	0	5	320	26	191	0	1	0	0	88	4836	257	0	789	0	0
4	0	100	0	0	0	0	0	0	0	37	0	0	0	0	0	0	0	643	0	0	0	0	0
5	1100	7317	283	381	1073	505	116	477	0	557	4	26	222	53	0	62	17719	17719	9244	1913	3538	9630	1050
6	150	1433	962	0	51	1386	0	74	118	118	0	26	0	601	0	0	1050	10039	789	9630	359	26375	2724
7	3708	417	0	0	0	0	2620	1669	431	0	0	0	0	0	1129	0	26375	2724	0	0	0	0	0

Decision Tree: Confusion Matrices																							
Train								Test								Full							
Actual	1	2	3	4	5	6	7	Actual	1	2	3	4	5	6	7	Actual	1	2	3	4	5	6	7
	26080	2767	29	0	422	73	1256		10515	1725	16	0	184	44	642		170341	27382	245	0	2942	709	10221
	4085	32205	954	30	2200	1071	413		2371	13123	433	15	996	454	162		363192	13881	6459	250	16084	7515	2793
	1	15	4169	319	46	619	0		0	20	1634	144	35	382	0		6	311	26805	2538	462	5632	0
	0	0	1	394	0	2	0		0	0	8	155	0	7	0		0	0	134	2517	0	96	0
	0	1	4	0	1364	4	0		10	73	18	0	478	9	0		115	945	188	0	8085	147	13
	0	9	185	100	19	2198	0		4	22	165	61	10	814	0		38	378	2406	859	168	13518	0
Predicted								Predicted								Predicted							
1	26080	4085	1	0	0	27	2935	1	10515	2371	0	0	0	4	66	1	170341	363192	1107	102	0	0	19279
2	2767	32205	15	1	9	2	0	2	1725	13123	20	73	22	7	7	0	27382	13881	102	0	22	0	0
3	29	954	4169	319	185	0	0	0	16	433	1634	18	165	0	0	0	245	6459	0	0	0	0	0
4	0	30	319	46	0	100	1	0	0	15	144	8	61	10	0	0	0	250	859	0	0	0	0
5	422	2200	46	0	19	0	0	0	184	996	35	0	10	0	0	0	0	16084	168	0	0	0	0
6	73	1071	619	2	4	0	0	0	44	454	382	9	0	0	0	0	709	7515	147	13	0	0	0
7	1256	413	0	0	0	0	2935	642	162	0	0	0	0	0	1198	0	10221	2793	0	0	0	0	0

Logistic Regression: accuracy=0.593, macro-F1=0.471, macro AUC=0.923
Decision Tree: accuracy=0.775, macro-F1=0.663, macro AUC=0.942

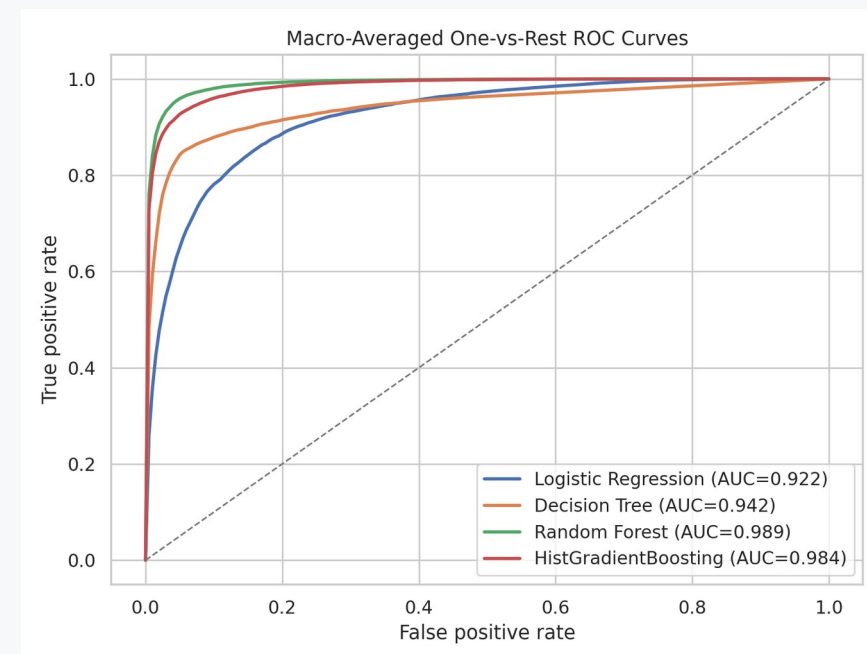
- Logistic regression reaches high macro AUC but weak class-level F1.
- The decision tree improves nonlinear separation but still lags tree ensembles.
- Both simple models are evaluated on train, test, and the full dataset after training.

Model Choice

Random Forest gives the best test macro-F1 and macro one-vs-rest AUC



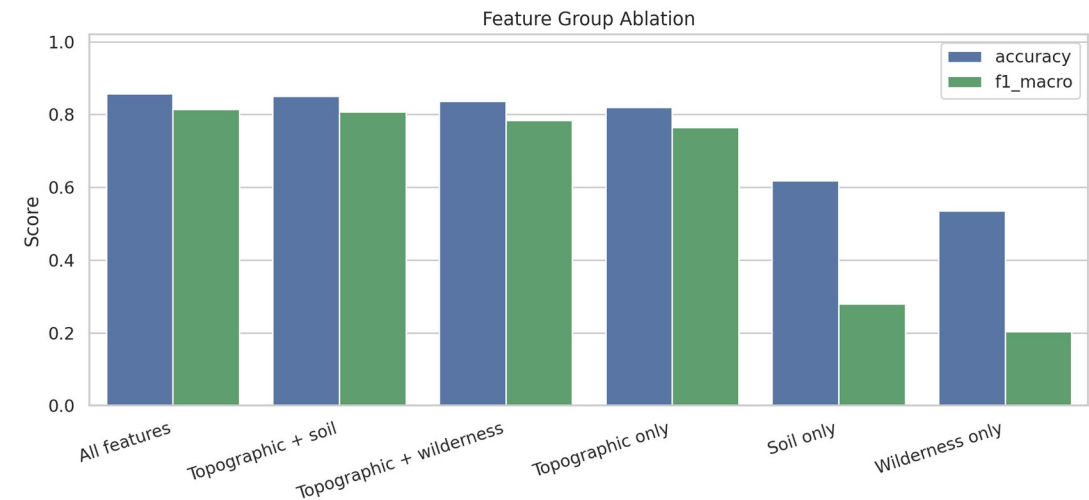
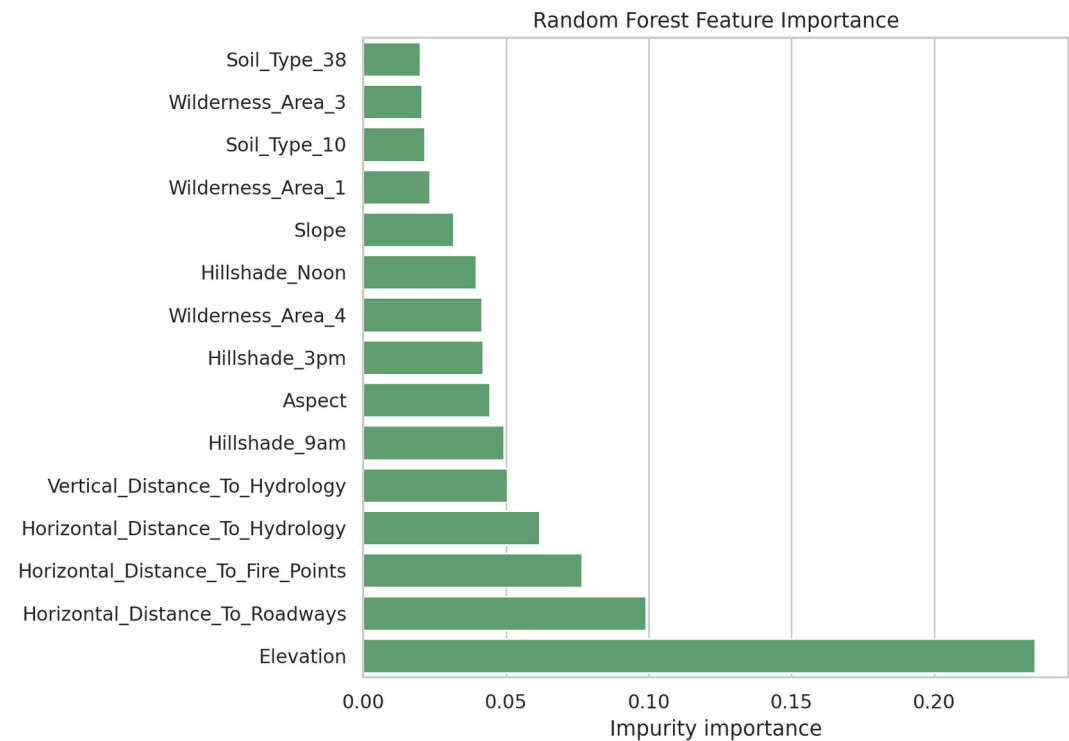
Logistic Regression: acc=0.593, macro-F1=0.471, AUC=0.923
Decision Tree: acc=0.775, macro-F1=0.663, AUC=0.942
Random Forest: acc=0.902, macro-F1=0.845, AUC=0.991
HistGradientBoosting: acc=0.869, macro-F1=0.831, AUC=0.985



- Cross-validation keeps the same ranking: Random Forest first, HistGradientBoosting second.
- Macro metrics matter because Cottonwood/Willow and Aspen are rare.

What Drives Cover Type?

Topography dominates, but soil indicators recover meaningful extra signal

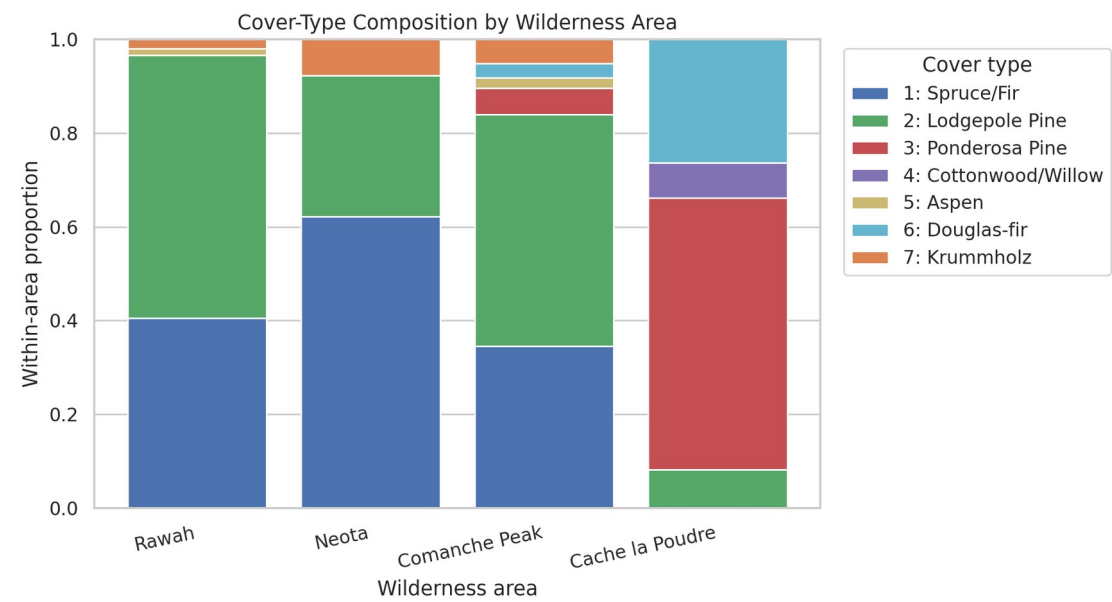


- 1. Elevation (0.235)
- 2. Horizontal_Distance_To_Roadways (0.099)
- 3. Horizontal_Distance_To_Fire_Points (0.077)
- 4. Horizontal_Distance_To_Hydrology (0.062)
- 5. Vertical_Distance_To_Hydrology (0.050)

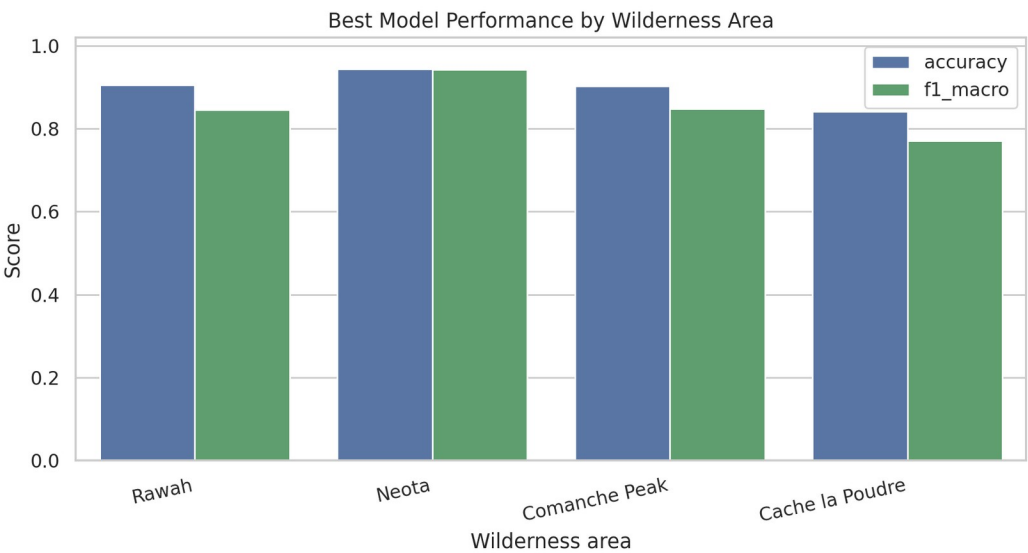
All features: F1=0.815
Topographic + soil: F1=0.807
Topographic + wilderness: F1=0.784
Topographic only: F1=0.764

Wilderness Area Extension

Regional composition and held-out performance explain where the model is easier or harder



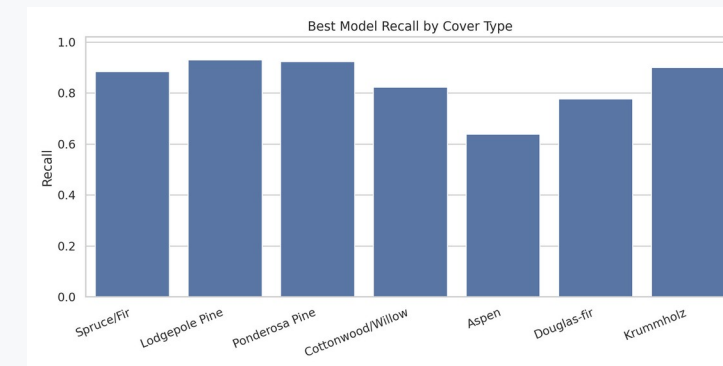
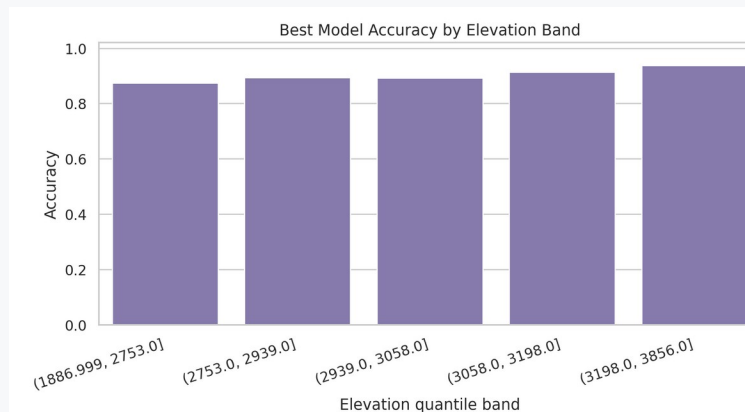
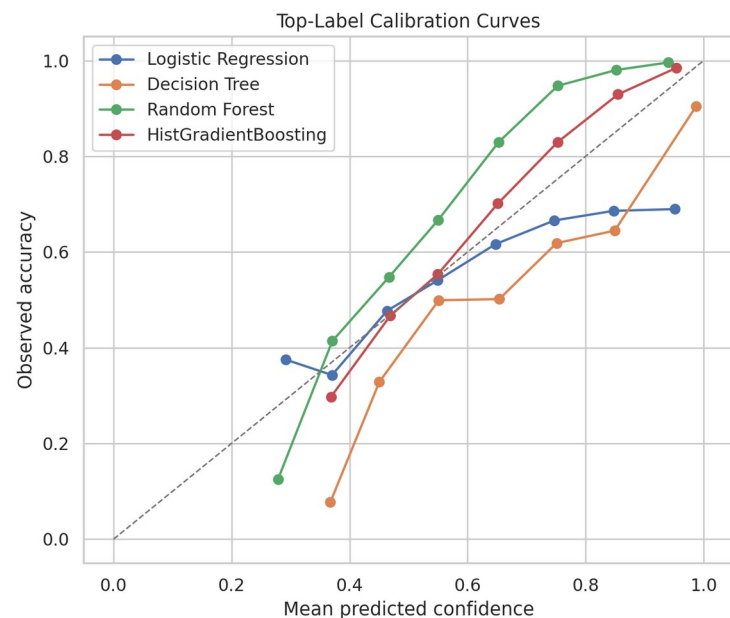
Rawah: 44.9%, dominant Lodgepole Pine (56.1%)
Neota: 5.1%, dominant Spruce/Fir (62.2%)
Comanche Peak: 43.6%, dominant Lodgepole Pine (49.4%)
Cache la Poudre: 6.4%, dominant Ponderosa Pine (58.0%)



- Hardest held-out area: Cache la Poudre, macro-F1 0.771.
- Regional evaluation checks robustness beyond aggregate test accuracy.
- Wilderness indicators alone are weak predictors, but the area split is useful for interpretation.

Calibration, Error Review, and Takeaways

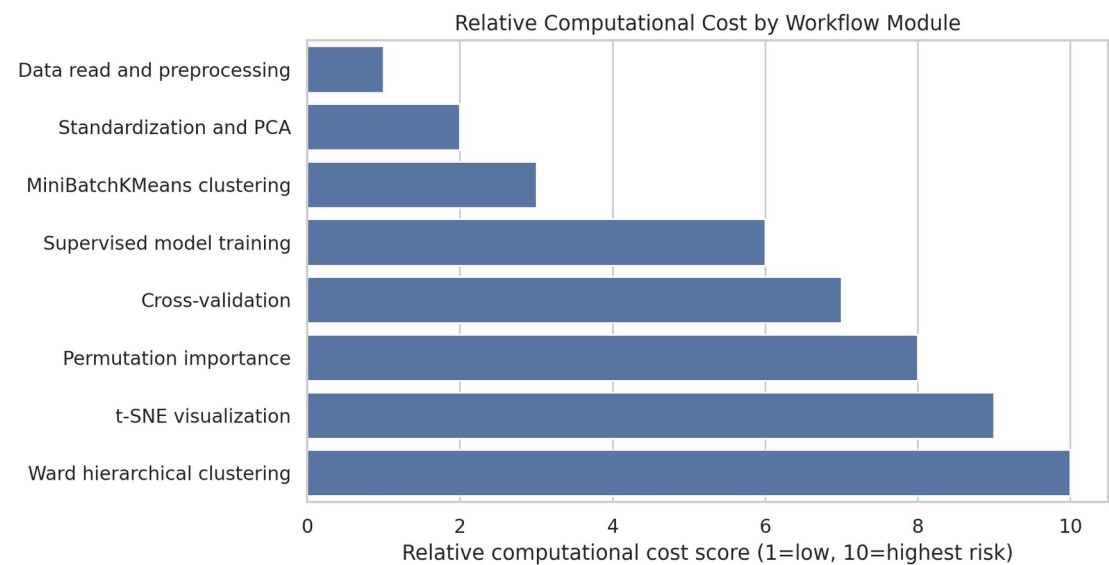
Accuracy is strong, but uncertainty and terrain bands still matter



- Best classifier: Random Forest with accuracy 0.902, macro-F1 0.845, macro AUC 0.991.
- Best calibration ECE: HistGradientBoosting at 0.047.
- Unsupervised clusters are useful for terrain/region exploration, but not for replacing supervised labels.
- Report macro metrics, class recall, calibration, elevation-band, and wilderness-area performance together.

Scalable Computation

Large-data workflow choices keep expensive steps reproducible



243.8 MB

naive int64 estimate

47.1 MB

optimized memory

t-SNE visualization: score 9
Ward hierarchical clustering: score 10
Permutation importance: score 8

- Memory reduction after downcast: 80.7%.
- Use full data for descriptive statistics, not for every diagnostic.
- Use stratified samples for t-SNE, Ward clustering, cross-validation, and permutation importance.
- Run PCA before distance-based visualization and clustering.
- Cache heavy stages with manifests; train scalable ensembles with parallelism.